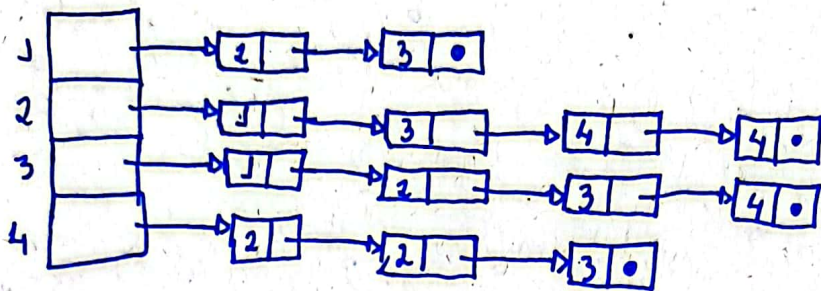


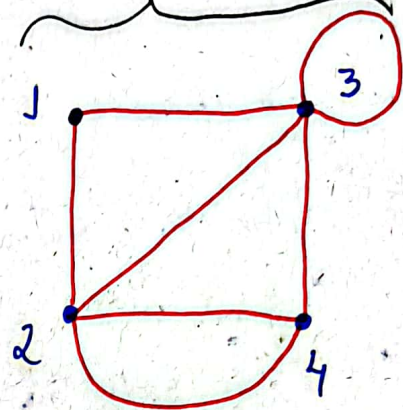
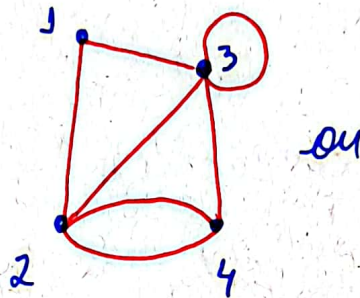
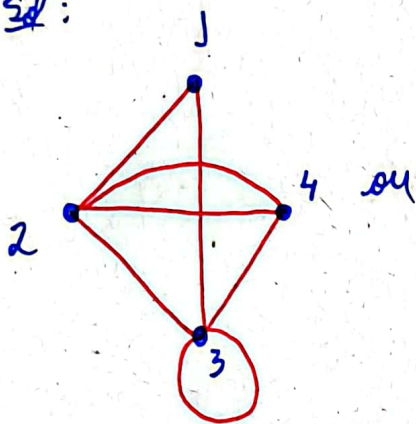
1) exercício 63 da pág. da lista de exercícios:

Ler o grafo não direcionado da lista de adjacências abaixo:

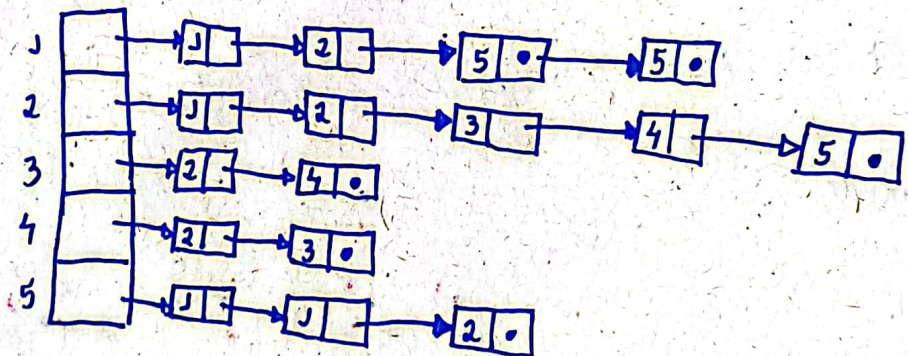
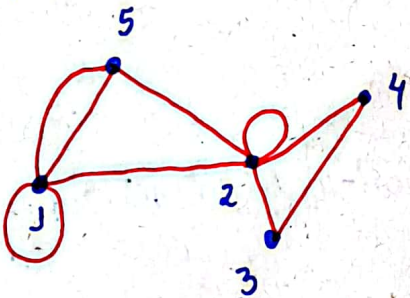


→ a resp. do livro
10, arrim
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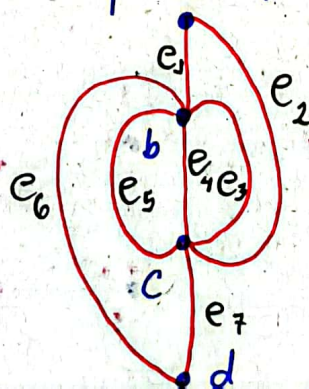
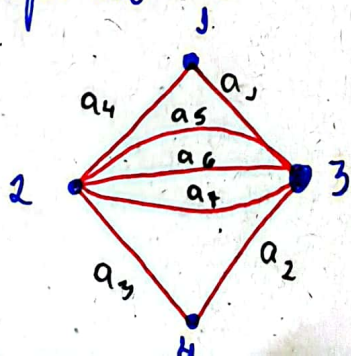
Sol:



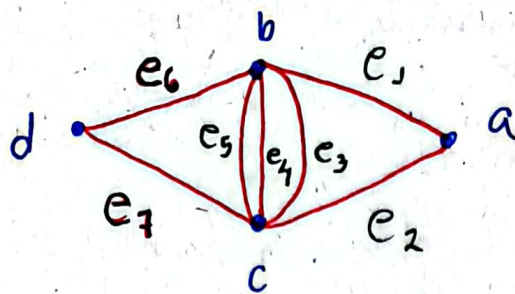
2) (exerc. 53) Ler a lista de adj. do grafo dado:



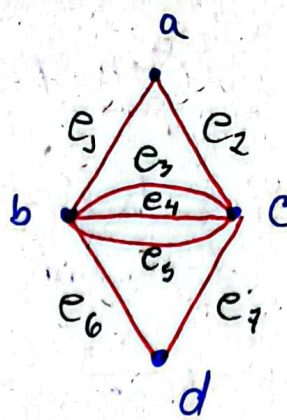
3) (exerc. 15) Verifique o isomorfismo (apresente as funções) dos grafos abaixo:



Sol:



$\Rightarrow$



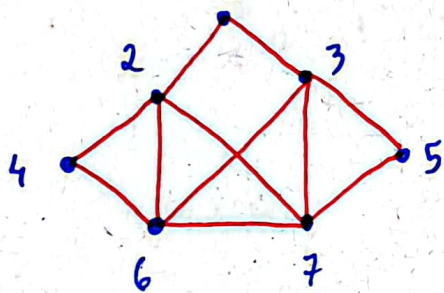
(2)

Logo, $f_1: 1 \rightarrow a, 2 \rightarrow b, 3 \rightarrow c, 4 \rightarrow d$

$f_2: a_1 \rightarrow e_2, a_2 \rightarrow e_7, a_3 \rightarrow e_6, a_4 \rightarrow e_1, a_5 \rightarrow e_3, a_6 \rightarrow e_4, a_7 \rightarrow e_5$

$\therefore$ são isomorfos.

4) (ex. 39) Escreva a matriz de adjacência do grafo



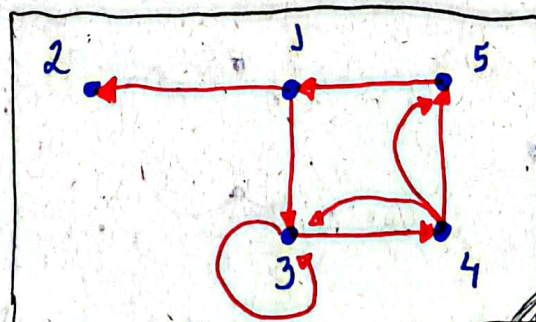
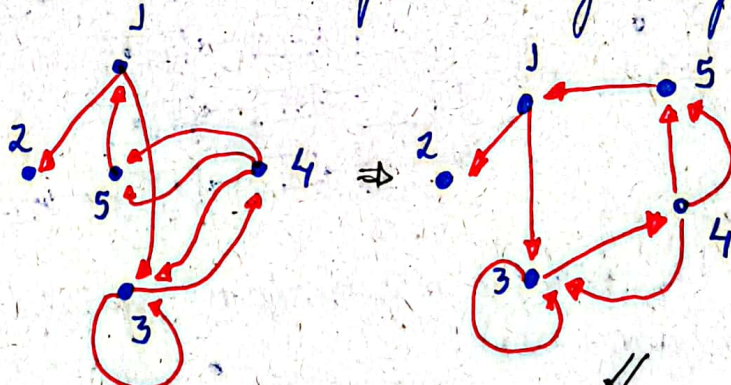
Sol:

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

5) (ex. 48) Desenhe o grafo direcionado da seguinte matriz de adjacência:

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

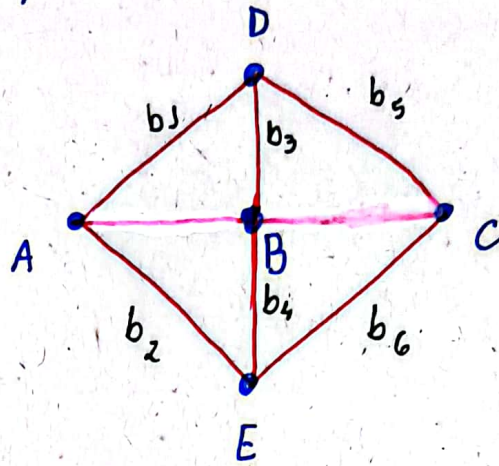
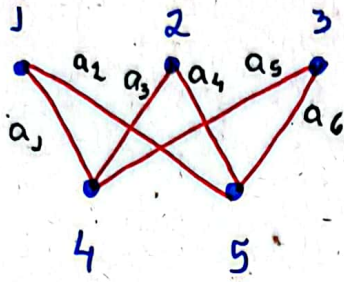
Sol:



6) Prove que o grafo $K_{2,3}$ é planar

(3)

Sol:



isomorfismo: $f_1: 1 \rightarrow A, 2 \rightarrow B, 3 \rightarrow C, 4 \rightarrow D, 5 \rightarrow E$

$f_2: a_1 \rightarrow b_1, a_2 \rightarrow b_2, a_3 \rightarrow b_3, a_4 \rightarrow b_4, a_5 \rightarrow b_5, a_6 \rightarrow b_6$

Como o 2º é planar e isomorfo ao 1º, temos que o 1º ($K_{2,3}$) é planar.